Instructions:

- **Due:** April 24, 11:59PM . Upload one **single PDF** to Blackboard containing your complete submission and upload all associated LaTeX and MATLAB source files.
- Each problem is worth $\dfrac{100}{\text{number of problems}}$ points.
- Start each problem on a new page.
- Label your PDF file exactly as

  `ENME408_S26_HW06_YourLastName.pdf`

- All work must be typeset in LaTeX. Handwritten or scanned submissions will not be accepted.
- Failure to follow these submission and formatting guidelines may result in a substantial point deduction or a score of zero.

Learning Objective: In many UAV applications, the full system state is not directly measured, and control laws must therefore be designed using estimated states reconstructed from available sensor outputs. The goal of this assignment is to develop insight into state estimation and output-feedback control for linear systems. In this homework, you will (i) analyze observability of linear systems, (ii) design an observer for state estimation, (iii) combine state feedback with state estimation to construct an observer-based output-feedback controller, and (iv) study the effect of observer pole placement on transient estimation and closed-loop behavior. Through analysis and simulation, you will observe how estimator dynamics interact with controller dynamics and how output feedback can recover the performance of full-state feedback when direct state measurements are unavailable.

Problem 1 (Observability of a Double-Integrator System)**.** Consider the system

$$\ddot{q} = u, \tag{1}$$

where $q \in \mathbb{R}$ is the position and $u \in \mathbb{R}$ is the control input.

1. Define the state vector by $x_1 = q$ and $x_2 = \dot{q}$. Write the system in the state-space form $\dot{x} = Ax + Bu$.
2. Suppose that only the position is measured, so that the output is $y = Cx$, where $y = q$. Determine the matrix C.
3. Form the observability matrix $\mathcal{O} = \begin{bmatrix} C \\ CA \end{bmatrix}$. Determine whether the pair (A, C) is observable.
4. Suppose instead that only the velocity is measured, so that $y = \dot{q}$. Determine the corresponding matrix C and test observability.
5. Briefly explain the physical meaning of observability in this example. In particular, discuss whether knowledge of only position measurements over time is sufficient to reconstruct the full state.

Problem 2 (Observer Design for a Double Integrator)**.** Consider again the double-integrator system

$$\dot{x} = Ax + Bu, \qquad y = Cx. \tag{2}$$

1. Consider the observer

   $$\dot{\hat{x}} = A\hat{x} + Bu + F(y - C\hat{x}), \tag{3}$$

   where $\hat{x}$ is the estimated state and $F \in \mathbb{R}^{2\times 1}$ is the observer gain. Define the estimation error $\tilde{x} = x - \hat{x}$. Show that the estimation error dynamics are governed by

   $$\dot{\tilde{x}} = (A - FC)\tilde{x}. \tag{4}$$

2. Let $F = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix}$. Compute the characteristic polynomial of $A - FC$.

3. Choose F so that the observer eigenvalues are located at $\lambda = -4$ and $\lambda = -5$. Determine the corresponding observer gain.

4. Simulate the plant and observer using $u(t) = 0$, initial conditions $x(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, and $\hat{x}(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Plot $x_1(t)$ and $\hat{x}_1(t)$ on one plot, and $x_2(t)$ and $\hat{x}_2(t)$ on another plot.

5. Plot the estimation error $\tilde{x}(t)$ and briefly describe the behavior of the observer.

Problem 3 (Observer-Based Output Feedback)**.** Consider the double-integrator system

$$\dot{x} = Ax + Bu, \qquad y = Cx. \tag{5}$$

1. Consider the state-feedback control law

$$u = Kx, \tag{6}$$

where $K = \begin{bmatrix} k_1 & k_2 \end{bmatrix}$. Derive the closed-loop system matrix $A + BK$.

2. Choose K so that the closed-loop eigenvalues are located at $\lambda = -2 \pm 2j$. Determine the corresponding gain K.

3. Since the full state is not measured, replace x in the control law by the estimate $\hat{x}$ and consider the output-feedback control law

$$u = K\hat{x}. \tag{7}$$

Write the combined plant-observer dynamics in terms of the augmented state $x_{\mathrm{a}} = \begin{bmatrix} x \\ \tilde{x} \end{bmatrix}$.

4. Show that the augmented dynamics can be written in block triangular form and identify its eigenvalues.

5. Explain why the controller eigenvalues and observer eigenvalues can be designed independently for this problem.

6. Use the controller gain from the previous part and the observer gain from the previous Problem. Simulate the closed-loop system with initial conditions $x(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\hat{x}(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Plot $y(t)$ and briefly describe the closed-loop response.

7. Compare this response with the response that would be obtained using full-state feedback $u = Kx$. Briefly comment on the effect of estimation transients.

Problem 4 (Effect of Observer Speed on Output-Feedback Performance)**.** Consider the observer-based controller from the previous problem.

1. Design a second observer whose eigenvalues are located at $\lambda = -10$ and $\lambda = -12$. Determine the corresponding observer gain.

2. Simulate the closed-loop system using both observer designs:

   (a) observer eigenvalues at -4 and -5,

   (b) observer eigenvalues at -10 and -12.

   Use the same controller gain and the same initial conditions as in Problem 3.

3. On the same figure, plot the output $y(t)$ for both cases. On a separate figure, plot the estimation error norm $\|\tilde{x}(t)\|$.

4. Compare the two responses and comment on how faster observer eigenvalues affect
   (a) estimation convergence,
   (b) output transient behavior,
   (c) possible sensitivity to measurement noise.
5. In practice, why is it not always desirable to place the observer eigenvalues arbitrarily far to the left in the complex plane?